ATMA RAM SANATAN DHARMA COLLEGE

(UNIVERSITY OF DELHI)

Practical File of Computer Networks

Submitted By :

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Course: B.Sc (Hons.) Computer Science

Semester : 4

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Ques-1. To study basic network command and network configuration commands

Solution:

```
C:\Users\gs070>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::37b6:c157:4acc:2b81%17
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2405:201:4035:f818:fe4d:65cc:2c92:effc
   Temporary IPv6 Address. . . . . . : 2405:201:4035:f818:1443:11c8:e7ed:5da0
   Link-local IPv6 Address . . . . . : fe80::46d3:f6f4:f9e7:11c3%15
   IPv4 Address. . . . . . . . . . . : 192.168.29.210
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1a82:8cff:fee7:fdc7%15
                                       192.168.29.1

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
```

```
C:\Users\gs070>ping google.com

Pinging google.com [2404:6800:4002:81e::200e] with 32 bytes of data:
Reply from 2404:6800:4002:81e::200e: time=6ms
Reply from 2404:6800:4002:81e::200e: time=6ms
Reply from 2404:6800:4002:81e::200e: time=7ms
Reply from 2404:6800:4002:81e::200e: time=7ms

Ping statistics for 2404:6800:4002:81e::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 7ms, Average = 6ms
```

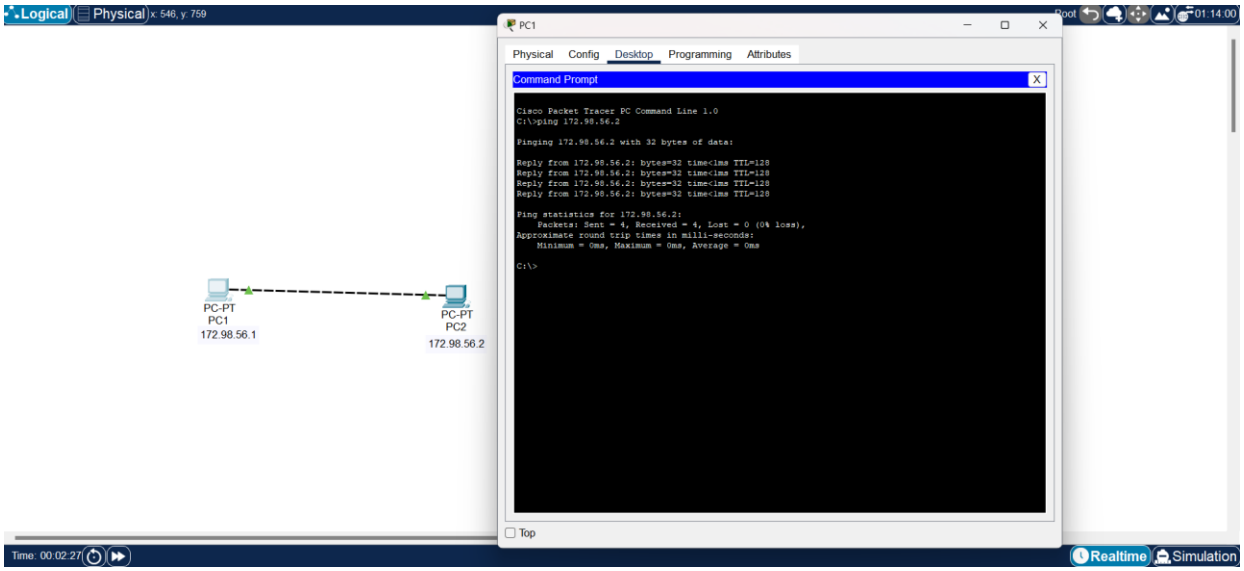
```
C:\Users\gs070>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:49671        gourav_vats:49672      ESTABLISHED
  TCP    127.0.0.1:49672        gourav_vats:49671      ESTABLISHED
  TCP    127.0.0.1:49673        gourav_vats:49674      ESTABLISHED
  TCP    127.0.0.1:49674        gourav_vats:49673      ESTABLISHED
  TCP    192.168.29.210:49412   20.198.119.84:https    ESTABLISHED
  TCP    192.168.29.210:59704   20.198.118.190:https   ESTABLISHED
  TCP    192.168.29.210:59884   20.187.186.89:https    ESTABLISHED
  TCP    192.168.29.210:59886   20.187.186.89:https    ESTABLISHED
  TCP    192.168.29.210:59909   whatsapp-chatd-edge-shv-01-del1:5222  ESTABLISHED
  TCP    192.168.29.210:60016   152.195.38.76:http     TIME_WAIT
  TCP    192.168.29.210:60028   204.79.197.222:https   ESTABLISHED
  TCP    192.168.29.210:60029   152.195.38.76:http     ESTABLISHED
  TCP    192.168.29.210:60031   20.189.173.24:https    ESTABLISHED
  TCP    192.168.29.210:60032   204.79.197.254:https   ESTABLISHED
  TCP    192.168.29.210:60036   del03s15-in-f3:https   TIME_WAIT
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:59887  sb-in-f188:5228        ESTABLISHED
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60005  [2405:200:1630:1600::312c:8d71]:https  CLOSE_WAIT
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60009  [2603:1046:1400::1]:https  TIME_WAIT
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60013  [2405:200:1630:1600::312c:8d21]:https  CLOSE_WAIT
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60020  [2405:200:1630:1600::312c:8d0a]:https  CLOSE_WAIT
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60034  [2606:2800:147:120f:30c:1ba0:fc6:265a]:https  ESTABLISHED
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60037  del11s06-in-x03:https  TIME_WAIT
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60038  del11s07-in-x03:https  TIME_WAIT
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60039  rg-in-f94:https        TIME_WAIT
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60040  whatsapp-cdn6-shv-01-del2:https  ESTABLISHED
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60041  whatsapp-cdn6-shv-01-del2:https  ESTABLISHED
  TCP    [2405:201:4035:f818:1443:11c8:e7ed:5da0]:60042  whatsapp-cdn6-shv-01-del2:https  ESTABLISHED
```

```
C:\Users\gs070>tracert www.google.com

Tracing route to www.google.com [2404:6800:4002:81a::2004]
over a maximum of 30 hops:

  1     2 ms     1 ms     1 ms  2405:201:4035:f818:1a82:8cff:fee7:fdc7
  2     *        *        *     Request timed out.
  3     5 ms     5 ms     5 ms  2405:200:801:300::71
  4     *        *        *     Request timed out.
  5     7 ms     7 ms     5 ms  2001:4860:1:1::1ef4
  6     6 ms     6 ms     6 ms  2404:6800:8074::1
  7     7 ms     6 ms     6 ms  2001:4860:0:1::43b8
  8     5 ms     5 ms     5 ms  2001:4860:0:1::54fb
  9     7 ms     5 ms     5 ms  del11s15-in-x04.1e100.net [2404:6800:4002:81a::2004]

Trace complete.
```

```
C:\Users\gs070>hostname
gourav_vats
```

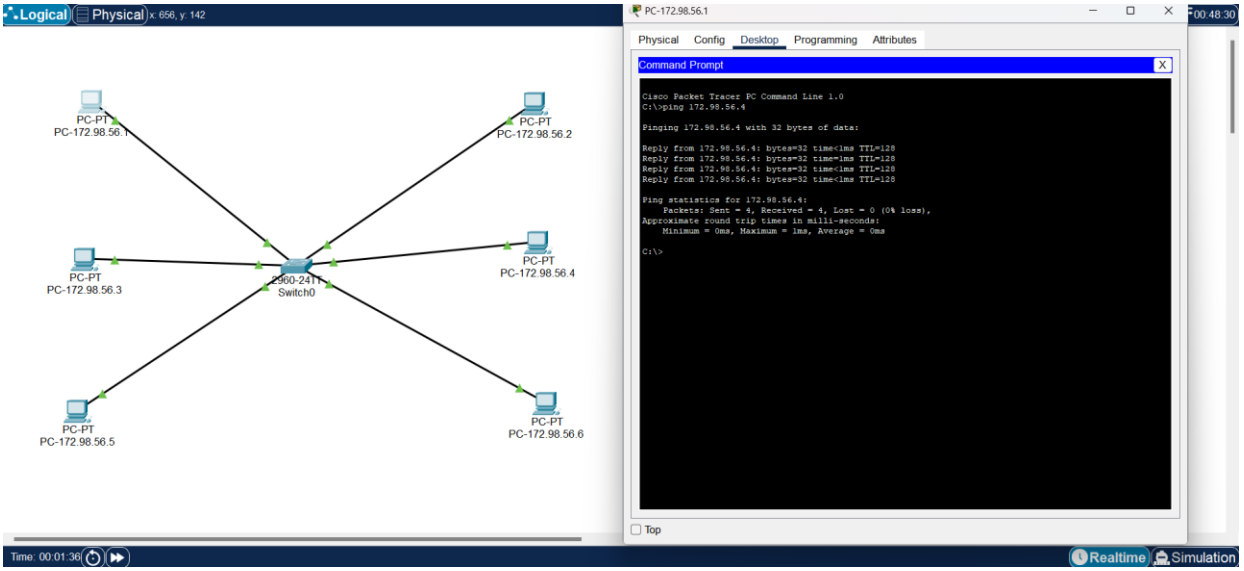
Ques-2. To study and perform PC to PC communication using ethernet

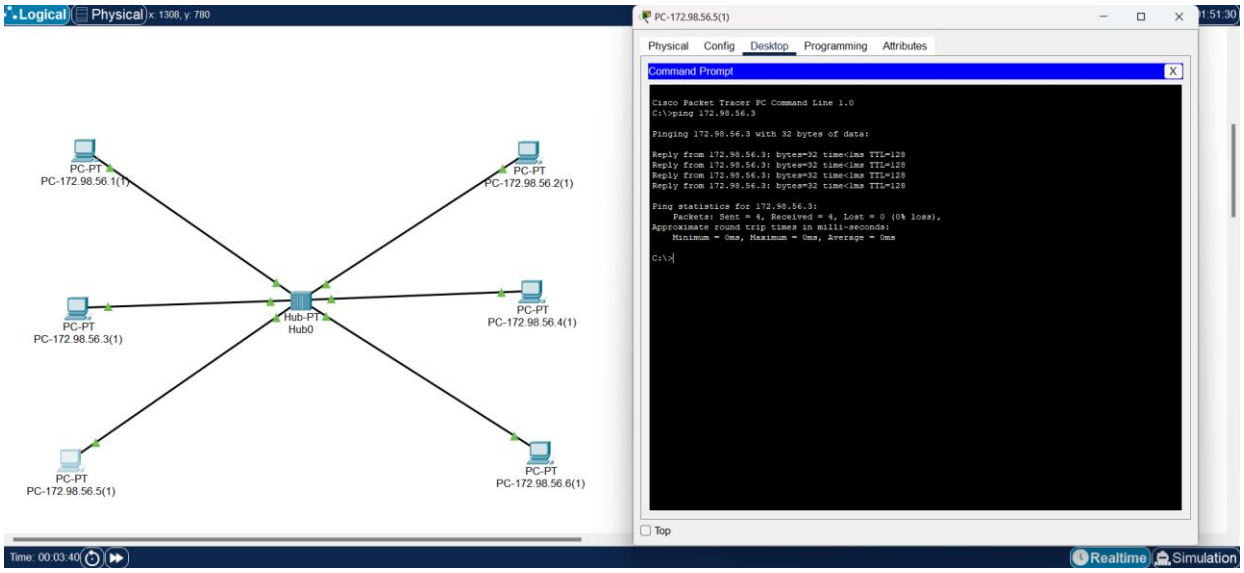
Solution:



Ques-3. To create Star topology using hub and switch

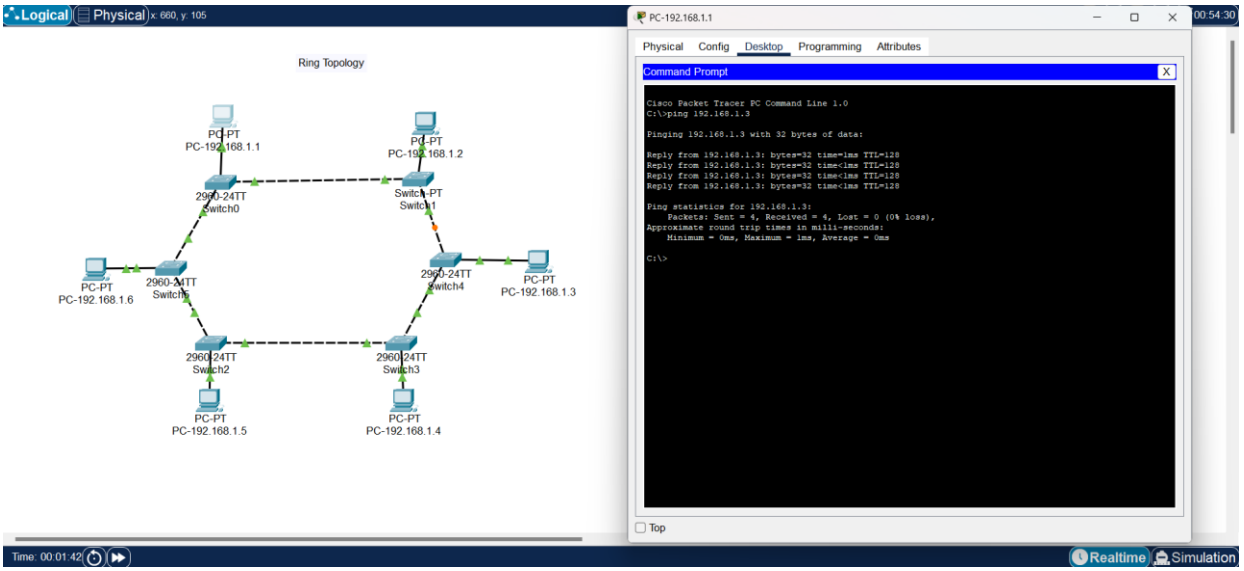
Solution:

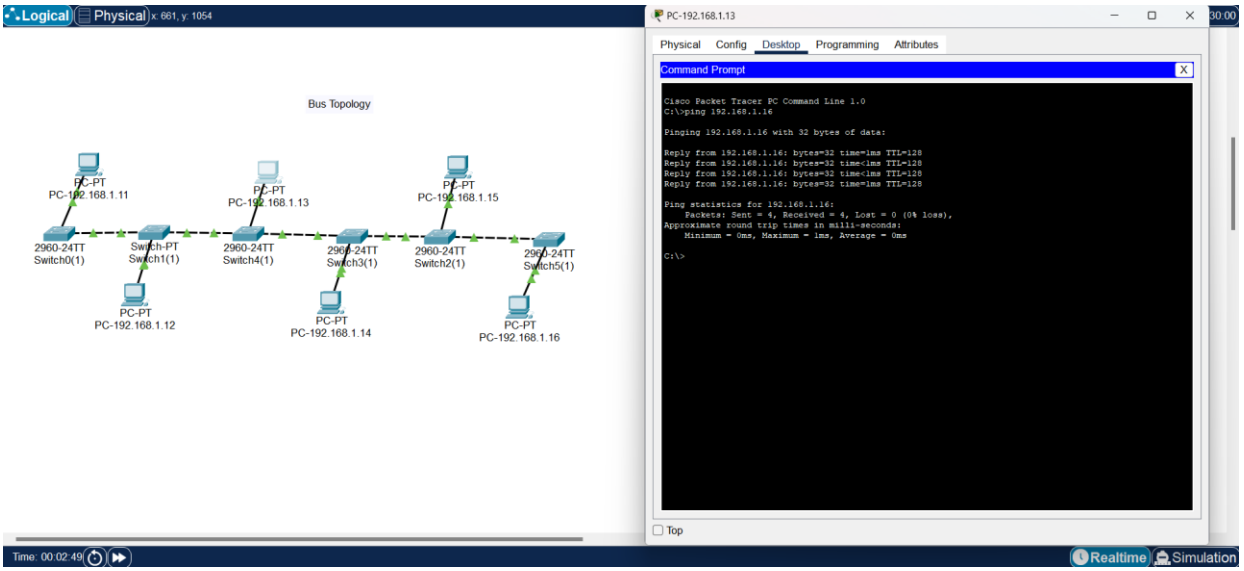


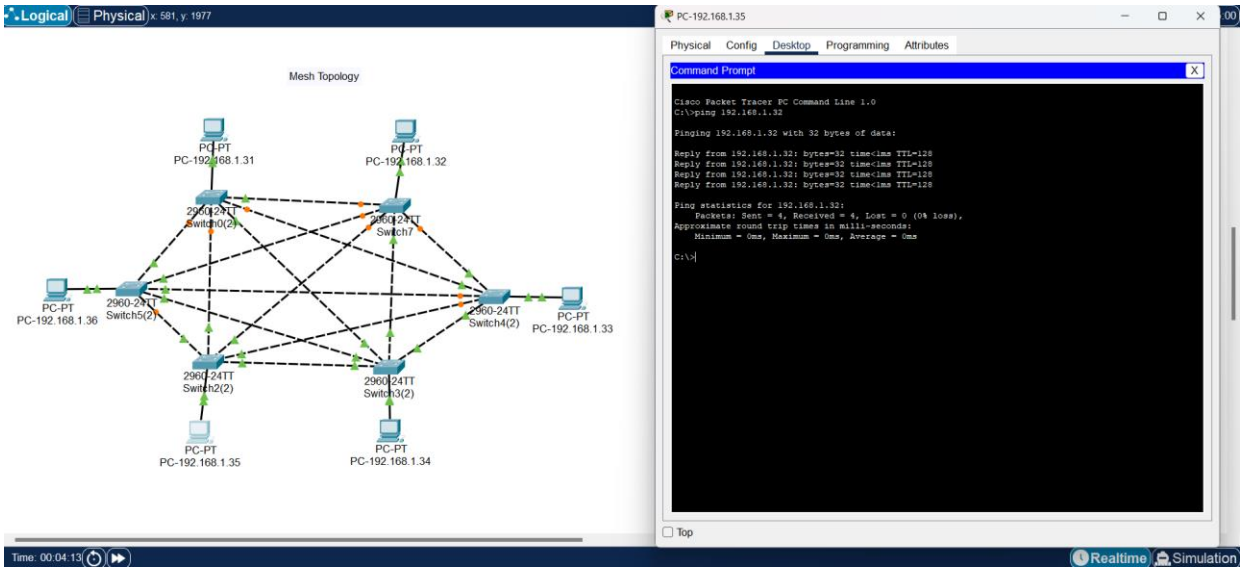


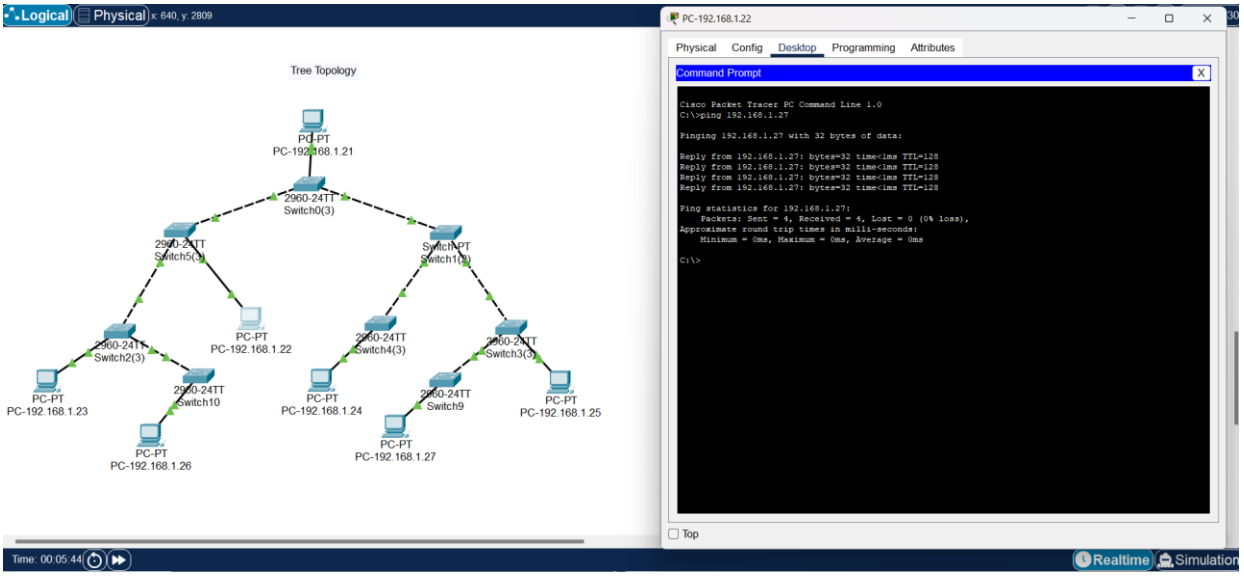
Ques-4. To create Bus, Ring, Tree, Hybrid, Mesh topologies

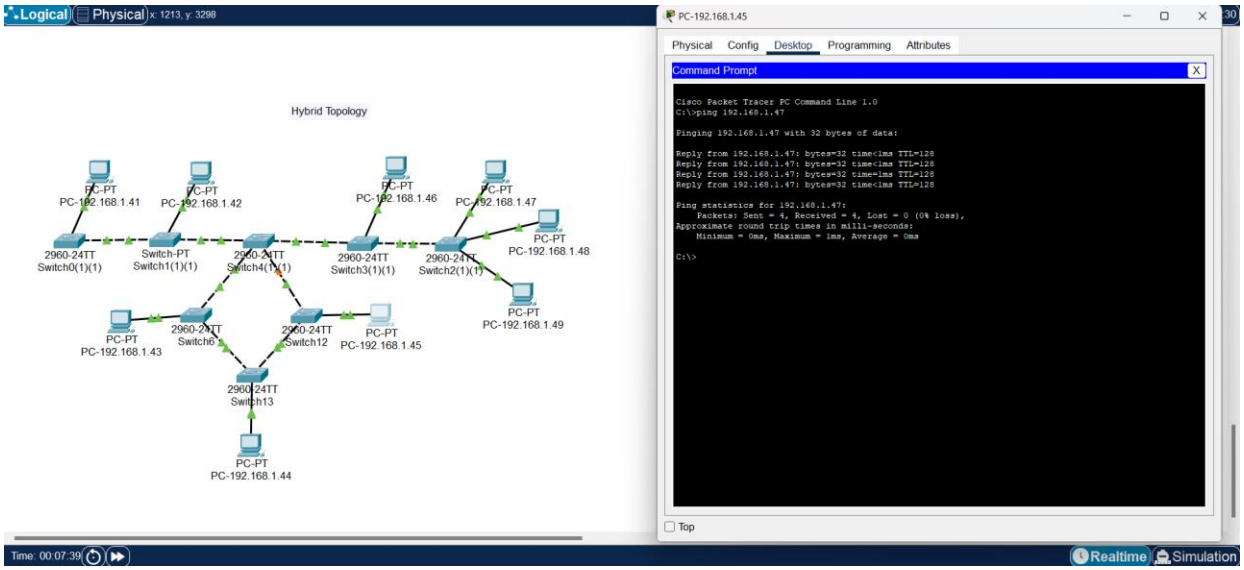
Solution:




Logical
Physical
x: 661, y: 1054
Bus Topology
PC-PT
PC-192.168.1.11
PC-PT
PC-192.168.1.13
PC-PT
PC-192.168.1.15
2960-24TT
Switch0(1)
Switch-PT
Switch1(1)
2960-24TT
Switch4(1)
2960-24TT
Switch3(1)
2960-24TT
Switch2(1)
2960-24TT
Switch5(1)
PC-PT
PC-192.168.1.12
PC-PT
PC-192.168.1.14
PC-PT
PC-192.168.1.16
PC-192.168.1.13
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.16
Pinging 192.168.1.16 with 32 bytes of data:
Reply from 192.168.1.16: bytes=32 time=1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time<1ms TTL=128
Reply from 192.168.1.16: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.1.16:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
Top
Time: 00:02:49
Realtime
Simulation

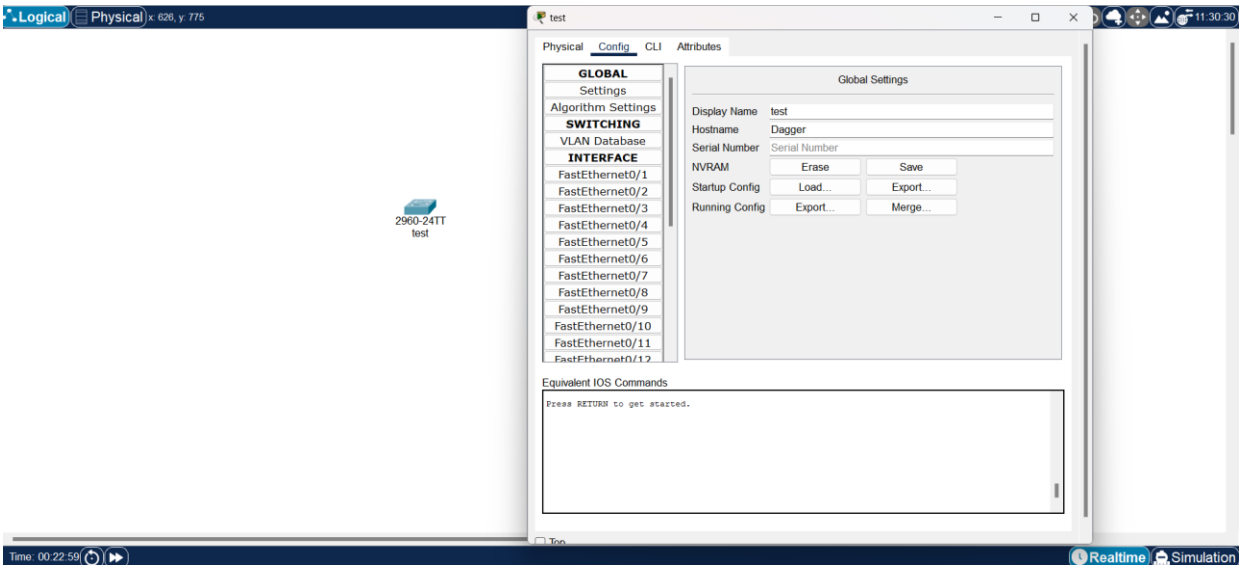

Logical
Physical
x: 581, y: 1977
Mesh Topology
PC-PT
PC-192.168.1.31
PC-PT
PC-192.168.1.32
2960-24TT
Switch0(2)
2960-24TT
Switch7
PC-PT
PC-192.168.1.36
2960-24TT
Switch5(2)
2960-24TT
Switch4(2)
PC-PT
PC-192.168.1.33
2960-24TT
Switch2(2)
2960-24TT
Switch3(2)
PC-PT
PC-192.168.1.35
PC-PT
PC-192.168.1.34
PC-192.168.1.35
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.32
Pinging 192.168.1.32 with 32 bytes of data:
Reply from 192.168.1.32: bytes=32 time<1ms TTL=128
Reply from 192.168.1.32: bytes=32 time<1ms TTL=128
Reply from 192.168.1.32: bytes=32 time<1ms TTL=128
Reply from 192.168.1.32: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.32:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Time: 00:04:13
Realtime
Simulation


Logical
Physical
x: 640, y: 2809
Tree Topology
PC-PT
PC-192.168.1.21
2960-24TT
Switch0(3)
2960-24TT
Switch5(3)
Switch-PT
Switch1(3)
PC-PT
PC-192.168.1.22
2960-24TT
Switch2(3)
2960-24TT
Switch4(3)
2960-24TT
Switch3(3)
PC-PT
PC-192.168.1.23
2960-24TT
Switch10
PC-PT
PC-192.168.1.24
2960-24TT
Switch9
PC-PT
PC-192.168.1.25
PC-PT
PC-192.168.1.26
PC-PT
PC-192.168.1.27
PC-192.168.1.22
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.27
Pinging 192.168.1.27 with 32 bytes of data:
Reply from 192.168.1.27: bytes=32 time<1ms TTL=128
Reply from 192.168.1.27: bytes=32 time<1ms TTL=128
Reply from 192.168.1.27: bytes=32 time<1ms TTL=128
Reply from 192.168.1.27: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.27:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Time: 00:05:44
Realtime
Simulation



Ques-5. Perform an initial Switch configuration

Solution:



test

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started.



Dagger>enable
Dagger#config t
Dagger#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Dagger(config)#banner motd #
Enter TEXT message.  End with the character '#'.
!!!!!  COOL SWITCH  !!!!!
#

Dagger(config)#enable passw
Dagger(config)#enable password test@enable
Dagger(config)#line con 0
Dagger(config-line)#password test@console
Dagger(config-line)#login
Dagger(config-line)#exit
Dagger(config)#exit
Dagger#
%SYS-5-CONFIG_I: Configured from console by console

Dagger#exit
```

Copy Paste

Top

test

Physical Config CLI Attributes

IOS Command Line Interface

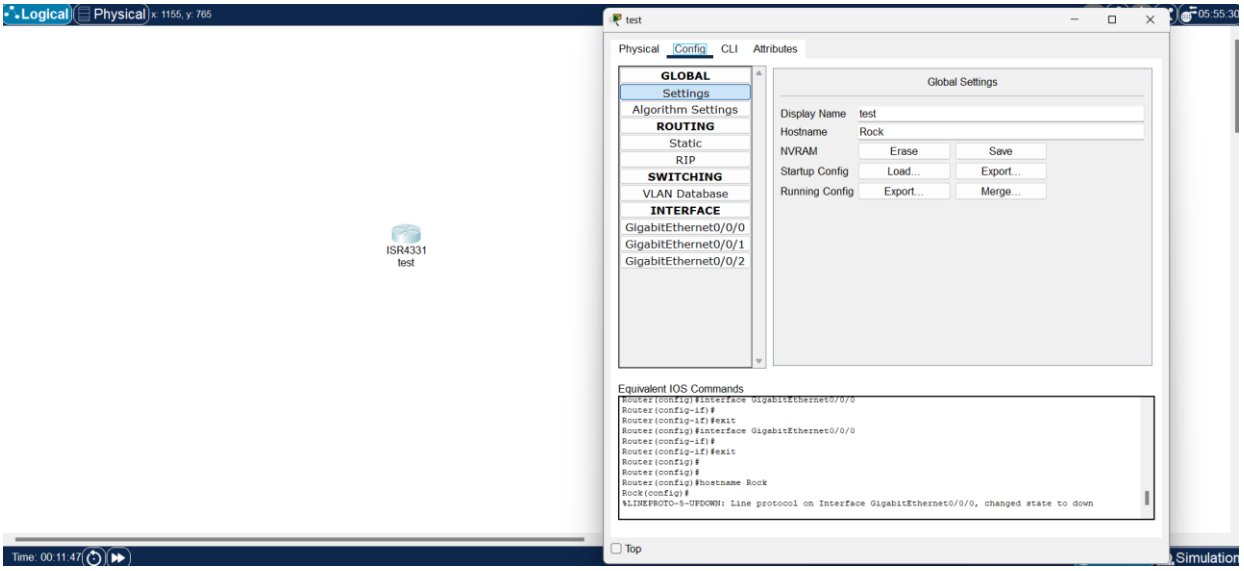
```
!!!!!  COOL SWITCH  !!!!!


User Access Verification

Password:
% Password:  timeout expired!



Press RETURN to get started!


!!!!!  COOL SWITCH  !!!!!


User Access Verification

Password:

Dagger>enable
Password:
Dagger#
Dagger#
Dagger#
```

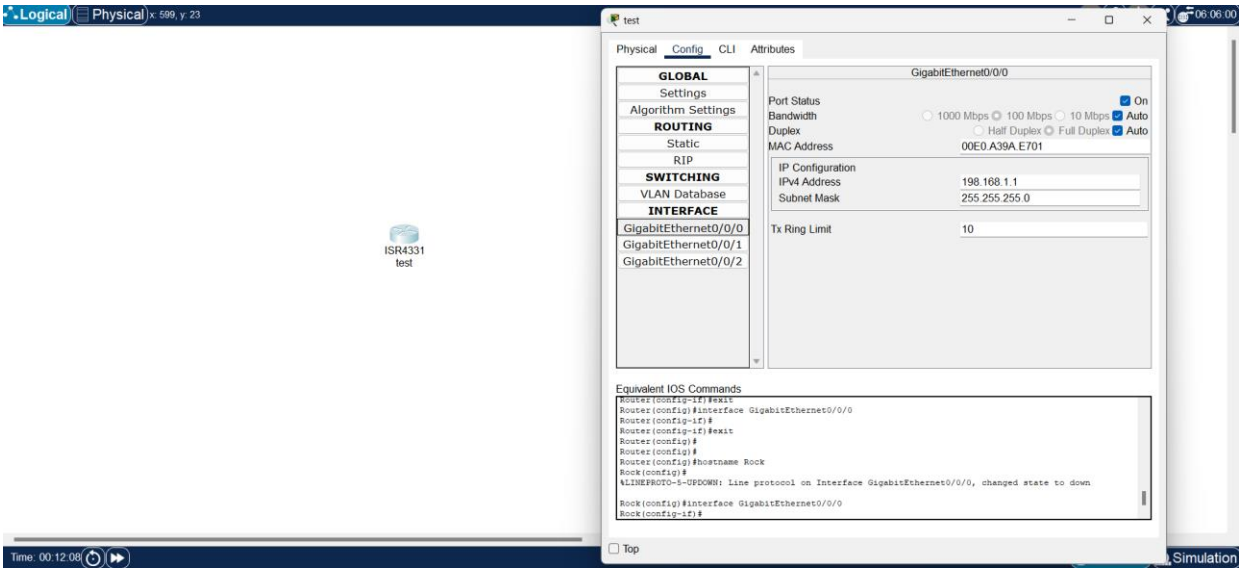
Copy Paste

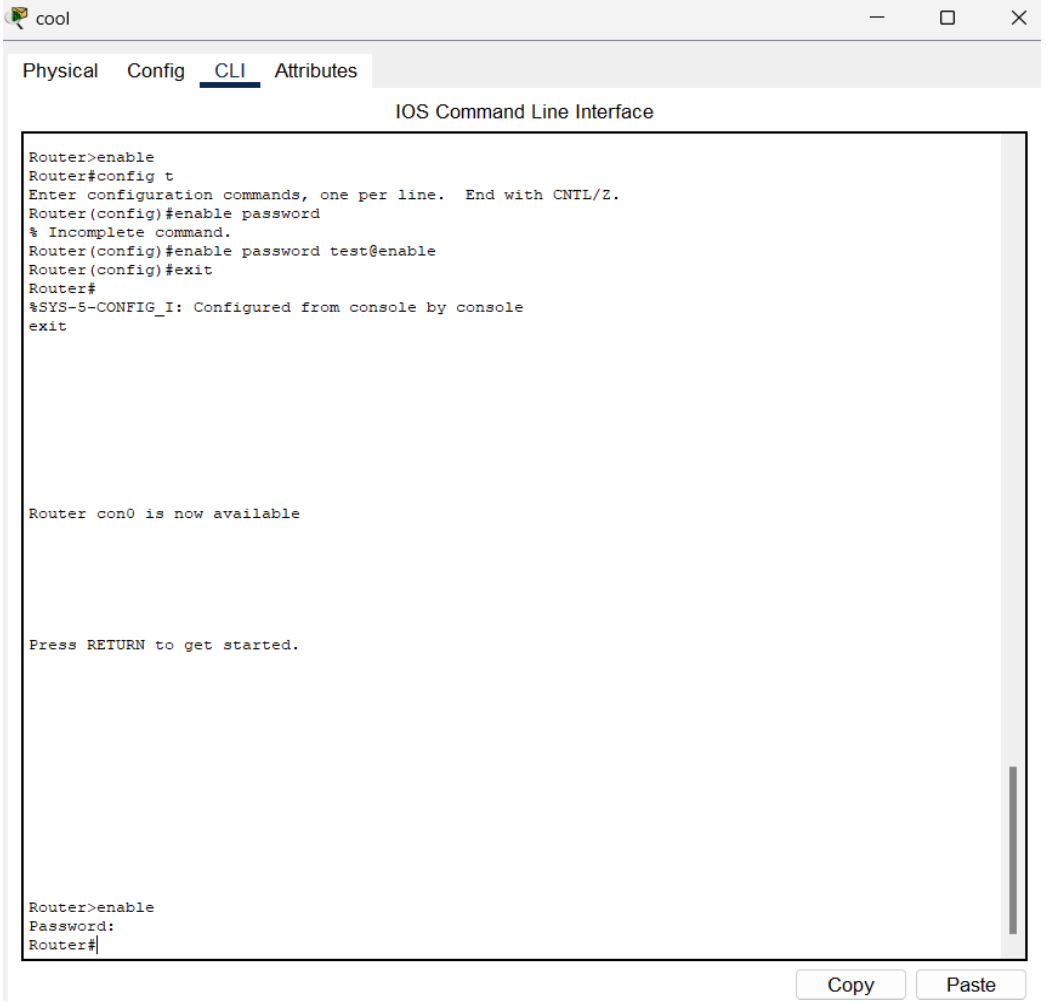
Top

Ques-6. Perform an initial Router configuration

Solution:

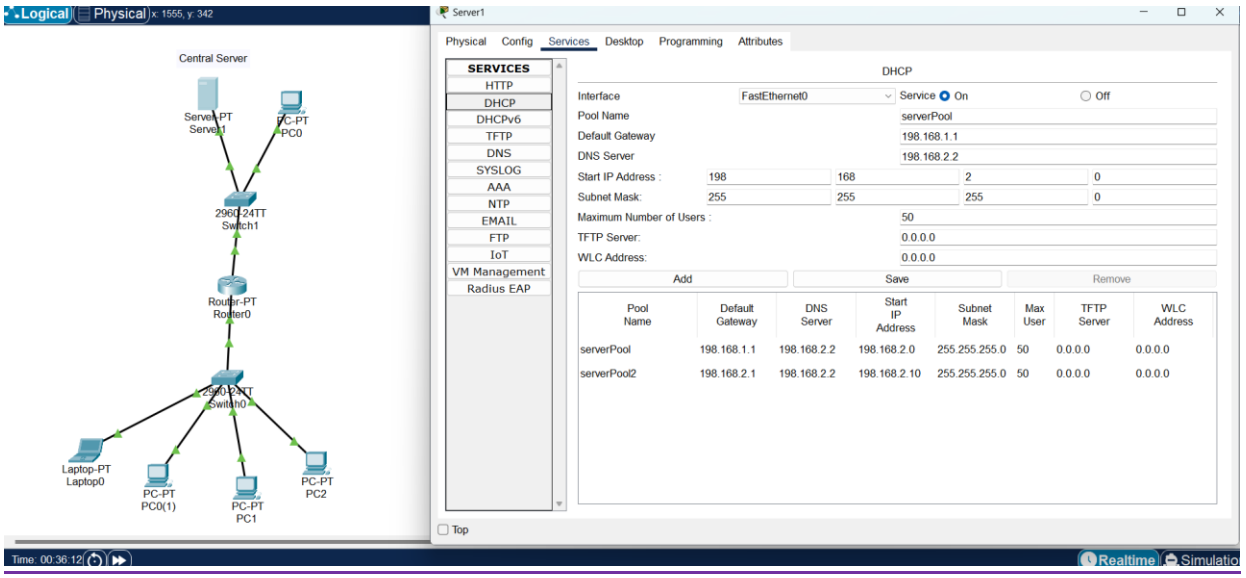


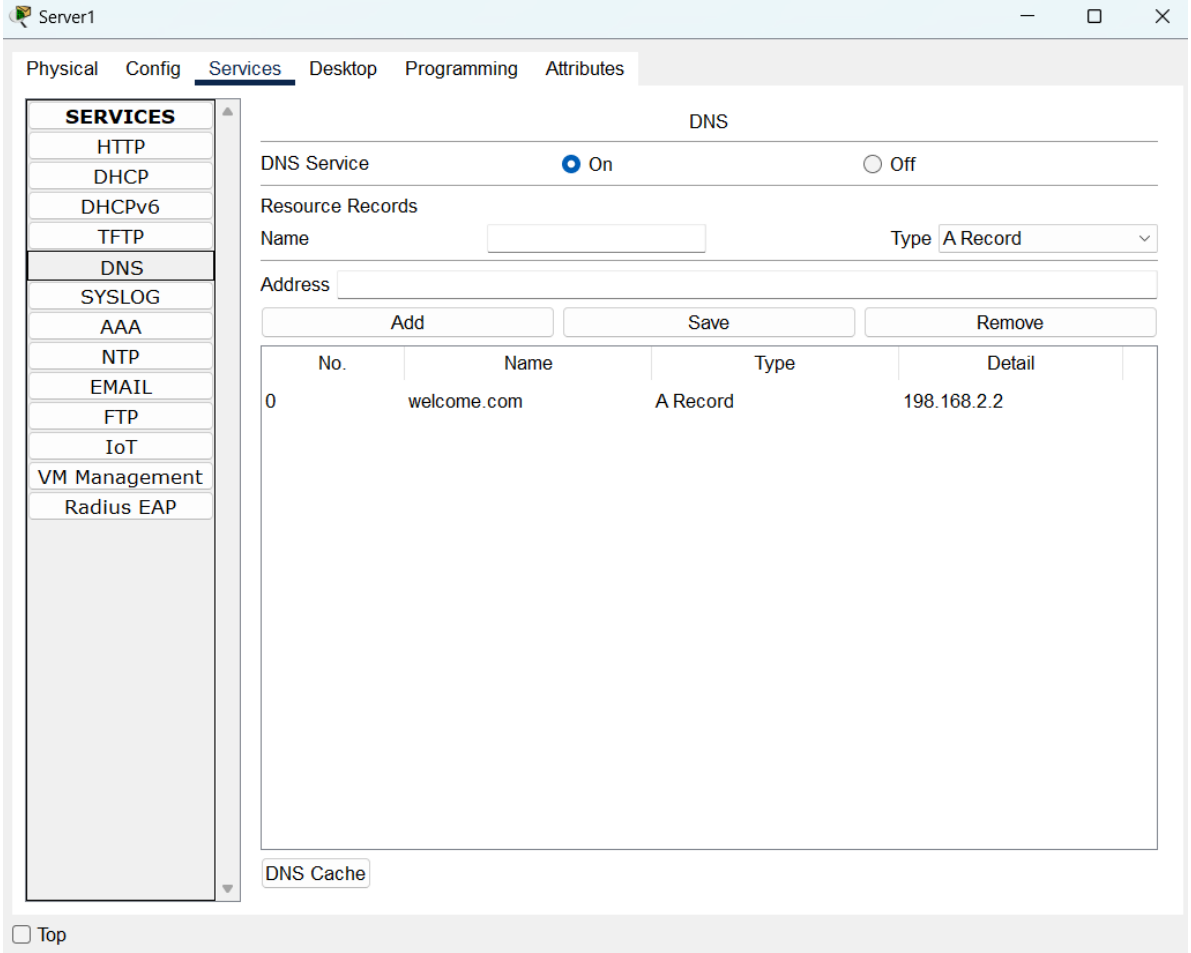


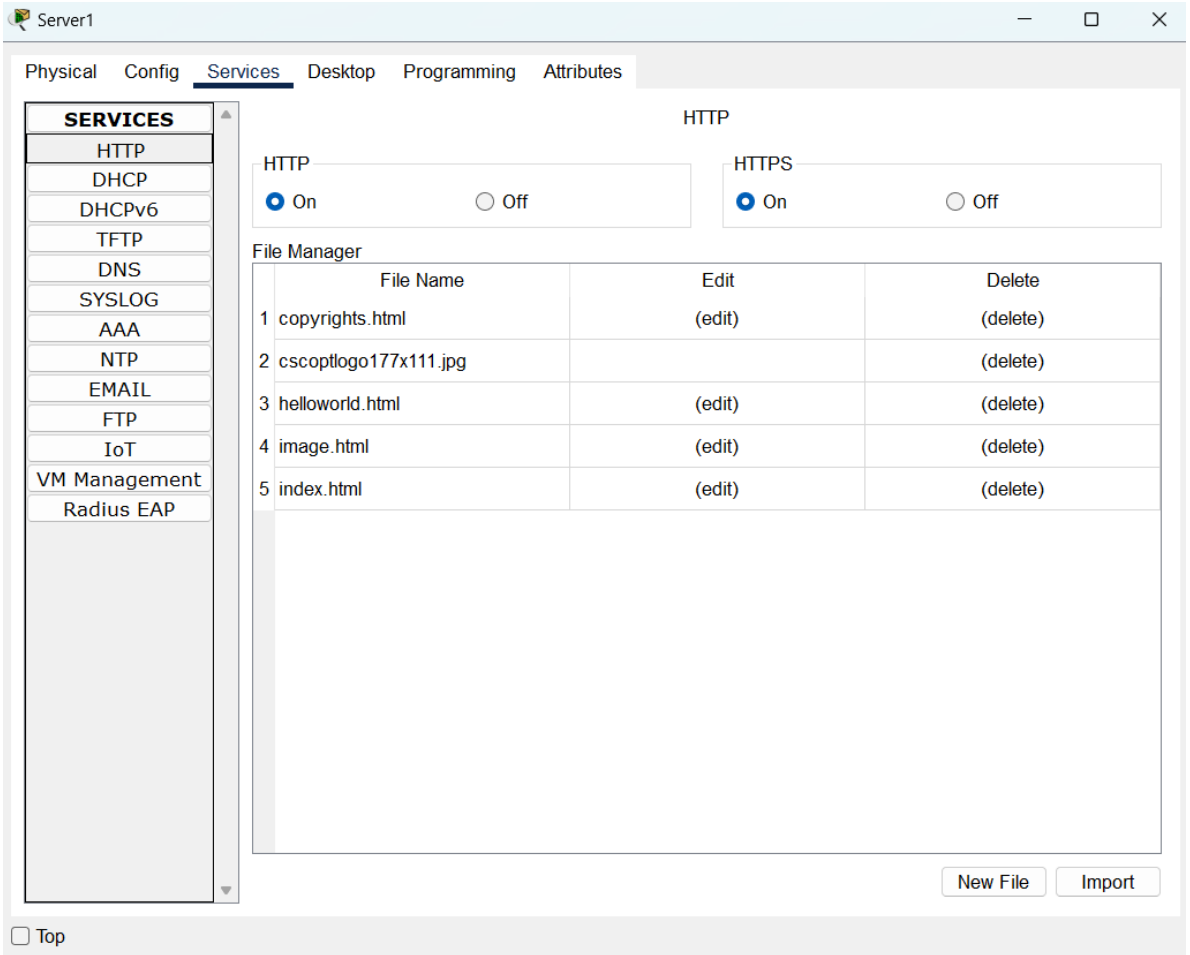


Ques-7. To implement Client Server Network

Solution:




Server1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0
welcome.com
A Record
198.168.2.2
DNS Cache
Top


Server1
Physical
Config
Services
Desktop
Programming
Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
HTTP
HTTP
On
Off
HTTPS
On
Off
File Manager
File Name
Edit
Delete
1 copyrights.html
(edit)
(delete)
2 cscoptlogo177x111.jpg
(delete)
3 helloworld.html
(edit)
(delete)
4 image.html
(edit)
(delete)
5 index.html
(edit)
(delete)
New File
Import
Top

PC0

Physical | Config | Desktop | Programming | Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

- DHCP
- Static

DHCP request successful.

IPv4 Address	198.168.2.11
Subnet Mask	255.255.255.0
Default Gateway	198.168.2.1
DNS Server	198.168.2.2

IPv6 Configuration

- Automatic
- Static

IPv6 Address	/
Link Local Address	FE80::202:4AFF:FE7B:3DCA
Default Gateway	
DNS Server	

802.1X

- Use 802.1X Security

Authentication	MD5
Username	
Password	

Top

PC0

Physical | Config | Desktop | Programming | Attributes

Web Browser

< > URL http://welcome.com Go Stop

Cisco Packet Tracer

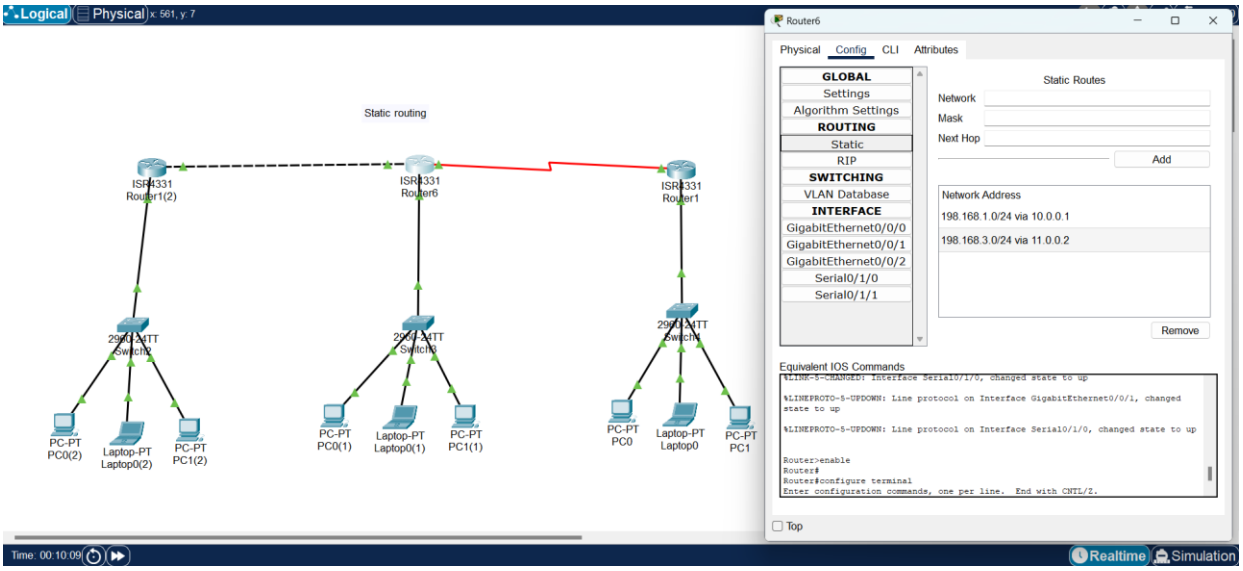
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.

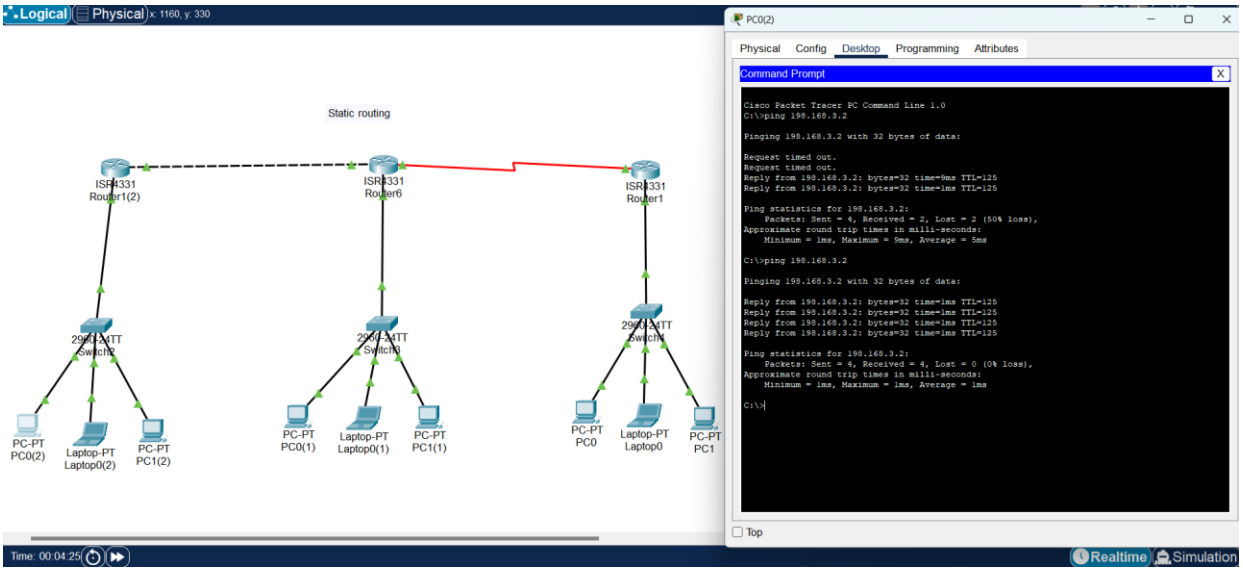
Quick Links:
A small page
Copyrights
Image page
Image

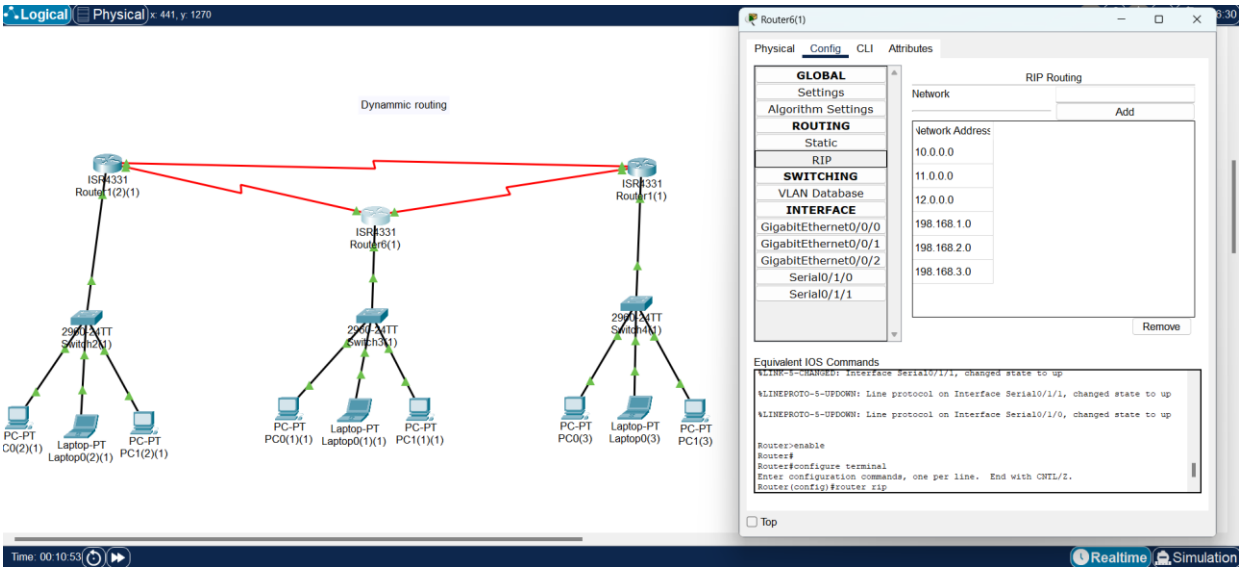
Top

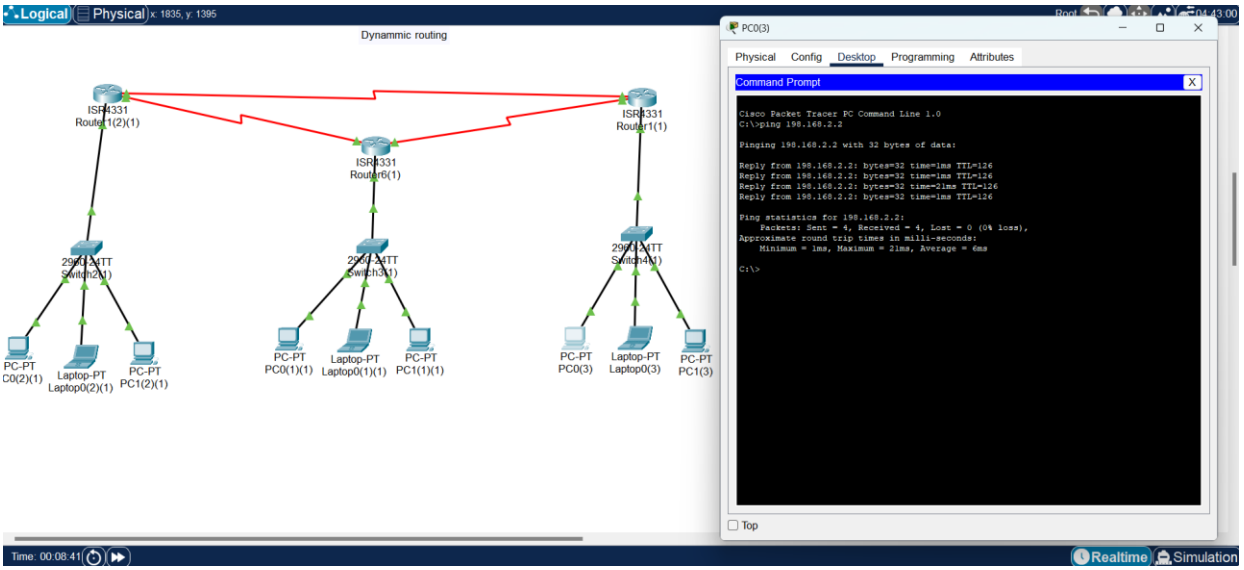
Ques-8. To implement connection between devices using router

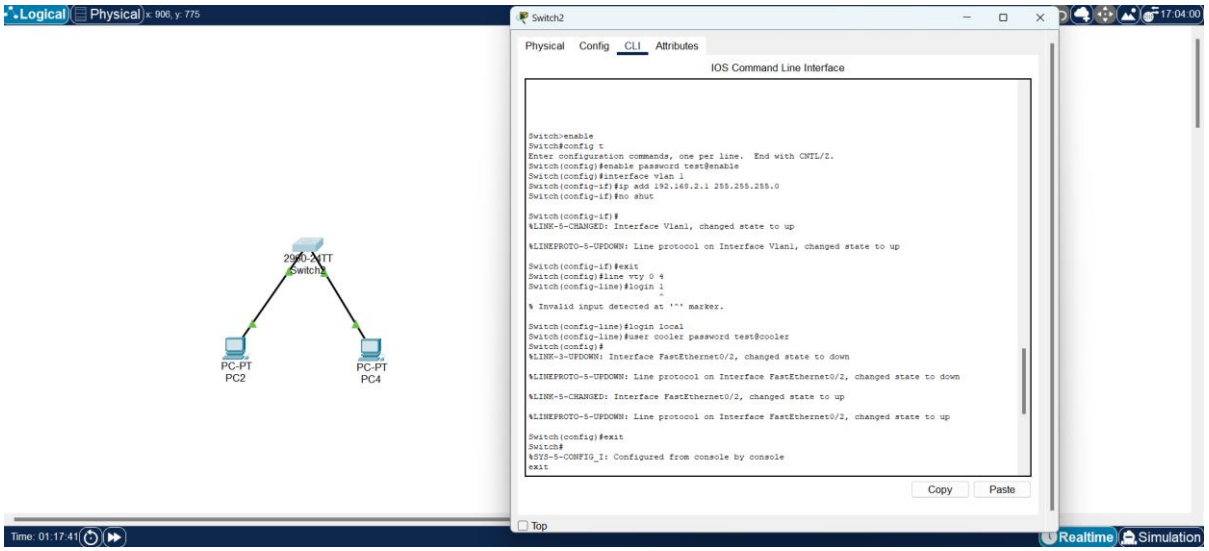
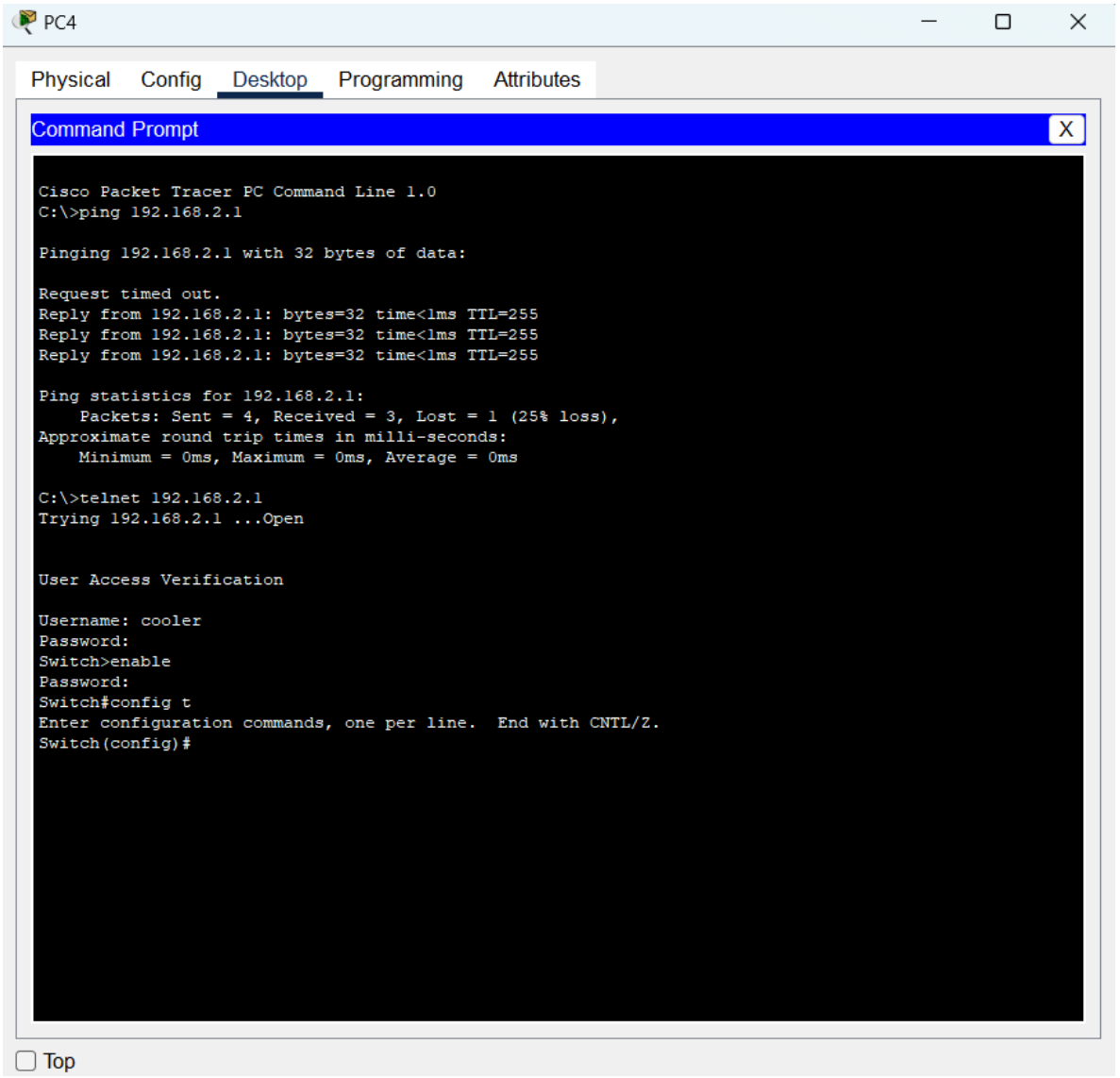
Solution:






Logical
Physical
x: 441, y: 1270
Dynammic routing
ISR4331
Router1(2)(1)
ISR4331
Router6(1)
ISR4331
Router1(1)
2960-24TT
Switch2(1)
2960-24TT
Switch3(1)
2960-24TT
Switch4(1)
PC-PT
C0(2)(1)
Laptop-PT
Laptop0(2)(1)
PC-PT
PC1(2)(1)
PC-PT
PC0(1)(1)
Laptop-PT
Laptop0(1)(1)
PC-PT
PC1(1)(1)
PC-PT
PC0(3)
Laptop-PT
Laptop0(3)
PC-PT
PC1(3)
Router6(1)
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0/0
GigabitEthernet0/0/1
GigabitEthernet0/0/2
Serial0/1/0
Serial0/1/1
RIP Routing
Network
Add
Network Address
10.0.0.0
11.0.0.0
12.0.0.0
198.168.1.0
198.168.2.0
198.168.3.0
Remove
Equivalent IOS Commands
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#router rip
Top
Time: 00:10:53
Realtime
Simulation


Logical
Physical
x: 1835, y: 1395
Dynammic routing
ISR4331
Router1(2)(1)
ISR4331
Router6(1)
ISR4331
Router1(1)
2960-24TT
Switch2(1)
2960-24TT
Switch3(1)
2960-24TT
Switch4(1)
PC-PT
C0(2)(1)
Laptop-PT
Laptop0(2)(1)
PC-PT
PC1(2)(1)
PC-PT
PC0(1)(1)
Laptop-PT
Laptop0(1)(1)
PC-PT
PC1(1)(1)
PC-PT
PC0(3)
Laptop-PT
Laptop0(3)
PC-PT
PC1(3)
PC0(3)
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 198.168.2.2
Pinging 198.168.2.2 with 32 bytes of data:
Reply from 198.168.2.2: bytes=32 time=1ms TTL=126
Reply from 198.168.2.2: bytes=32 time=1ms TTL=126
Reply from 198.168.2.2: bytes=32 time=21ms TTL=126
Reply from 198.168.2.2: bytes=32 time=1ms TTL=126
Ping statistics for 198.168.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 1ms, Maximum = 21ms, Average = 6ms
C:\>
Top
Time: 00:08:41
Realtime
Simulation

Ques-9. To perform remote desktop sharing within LAN connection

Solution: